Andrew Sarracini

Data Scientist · Applied AI

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SUMMARY

Data scientist who turns messy operational data into decisions people act on, using modern AI tooling to move several times faster. I started in clinical research at St. Joseph's Hospital, earned a Master's in Data Science, and now run all data and technology at Keyspire as sole technical lead: churn analysis, revenue-leak investigations, production pipelines, and an LLM agent measured by a ~130-check eval suite. I want to bring that to healthcare operations.

EXPERIENCE

Keyspire | **Head of Technology** Mar 2026 – Present

Promoted to succeed the departing Head of Technology; own all data, analytics, and AI work for a ~1,000-member education company.

- Took the Keyspire AI Assistant from prototype to paid launch in **~3.5 weeks**: an LLM coaching agent with RAG over a **337K-word** course library, persistent memory, and deterministic analysis tools, at **under 1¢ per reply**.
- Built the company's first churn baseline from raw billing data across two Stripe accounts: defined churn (people, not subscriptions, with a 30-day grace window), segmented by plan (**1.6%/mo** annual vs. **5–6.6%/mo** monthly), and showed early exits dominate (**169 of 356** churners left before a second payment).
- Root-caused a silent revenue leak: **49 renewals** ended with no card ever charged. Validated each case by hand to remove false positives, traced the cause to billing schedules created without a payment method, and flagged **41 more at risk (~$33K/yr)** before they billed.
- Made model behavior measurable: a **~130-check** live-model eval suite gates every behavior change. Cut consistently failing knowledge checks from **24 of 92 to 0** by tuning a retrieval relevance cutoff from measured score distributions.
- Run a nightly Python pipeline over **~1,000 subscriptions (~$40K MRR)**: **~975 automated updates** in **67 production runs**, Stripe reconciliation, and a nightly exceptions list the operations team works from, replacing **~20 hours/week** of manual work.
- Work as a scientist with agents: parallel AI coding agents, automated review, and CI let one person own analysis, modeling, and production code at once.

Keyspire | **Data Scientist** Jan 2025 – Mar 2026

Found that data quality was the company's real bottleneck and fixed it first.

- Led the HubSpot to GoHighLevel migration and cleansed **190K+ contact records** into a single source of truth.
- Built a two-way sync with the **900-member** Circle community and a **9,000-lead** outbound sequence; directed an external agency across **10+ projects**.

St. Louis Blues Hockey Club | **Data Scientist (Capstone Project)** May 2024 – Jul 2024

Built predictive models for a confidential classification task with an NHL organization.

- Raised balanced accuracy **more than 3x** on multi-source data with extreme class imbalance, missing values, and fuzzy name-matching: **28 engineered features**, regularization, and tuning across Gradient Boosting, Random Forest, and neural network models, validated against real-world outcomes.

St. Joseph's Hospital | **Research Data Analyst (part-time)** Sep 2018 – Dec 2019

Clinical research data work alongside psychiatric researchers.

- Co-developed study methodology and statistical analysis for **two psychiatric research papers**.
- Built a **150+ medication** reference dataset in REDCap for clinical research use.

EDUCATION

University of British Columbia Okanagan – Master of Data Science 2023–2024

Coursework: Machine Learning, Predictive Modelling, Bayesian Inference, Deep Learning, Neural Networks, Data Visualization, Database Management

McMaster University – Bachelor of Science (Honours) 2016–2021

SKILLS

Statistics & Modeling: Predictive modeling, Classification with class imbalance, Bayesian inference, Cohort & churn analysis, Feature engineering, Model evaluation
Machine Learning & Data: scikit-learn, XGBoost, PyTorch, Pandas, NumPy, PostgreSQL, REDCap
Languages: Python, SQL, R, TypeScript, Bash
Applied AI: LLM agents, RAG / vector search, Evals & guardrails, Agentic coding (Claude Code), MCP
Data Engineering: Production pipelines, Data cleansing, Stripe, CRM data, GitHub Actions, AWS, Google Cloud